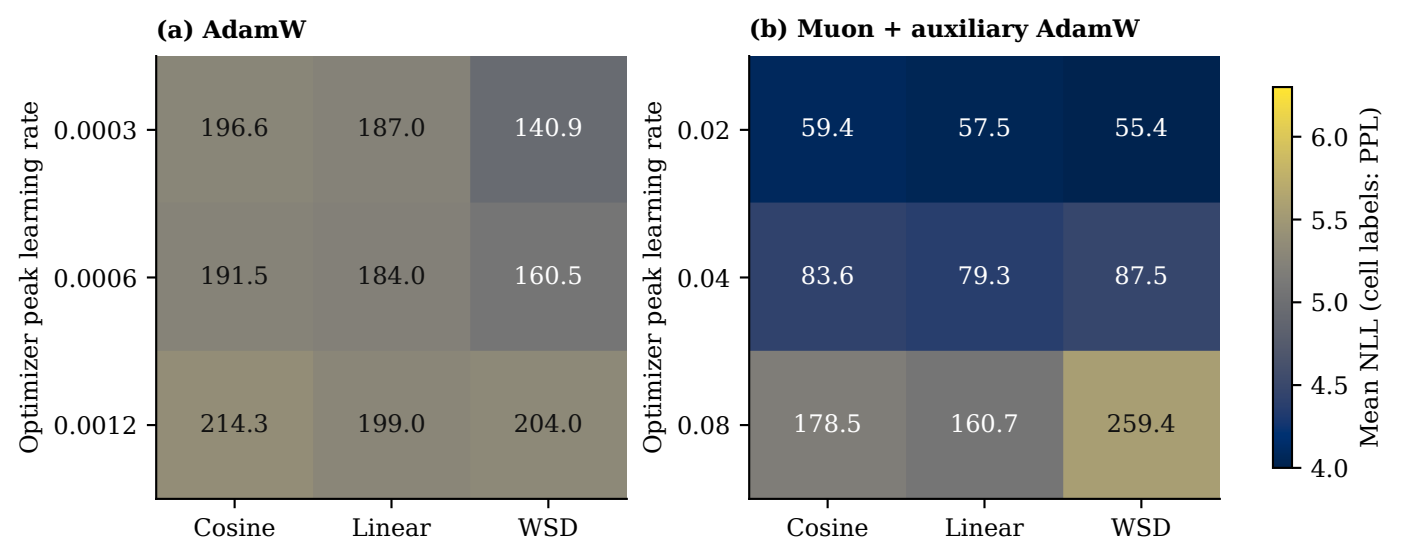
(a) AdamW
Optimizer peak learning rate
0.0003
0.0006
0.0012
196.6
187.0
140.9
191.5
184.0
160.5
214.3
199.0
204.0
Cosine
Linear
WSD
(b) Muon + auxiliary AdamW
Optimizer peak learning rate
0.02
0.04
0.08
59.4
57.5
55.4
83.6
79.3
87.5
178.5
160.7
259.4
Cosine
Linear
WSD
Mean NLL (cell labels: PPL)
6.0
5.5
5.0
4.5
4.0